Applied Data Mining

— Week 2—

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Dr. Hashim Safdar

Outline

- Data Objects and Attribute Types
- Basic Statistical Descriptions of Data

Data Objects

- Data sets are made up of data objects.
- A **data object** represents an entity.
- Examples:
  - sales database: customers, store items, sales
  - medical database: patients, treatments
  - university database: students, professors, courses
- Also called *samples , examples, instances, data points, objects, tuples*.
- Data objects are described by **attributes**.
- Database (rows -> data objects; columns ->attributes)

Attributes

- **Attribute** (or **dimensions, features, variables**): a data field, representing a characteristic or feature of a data object.
  - E.g., customer _ID, name, address
- A set of attributes used to describe a given object is called an attribute vector (or feature vector).
- The distribution of data involving one attribute (or variable) is called uni-variate.
- A bi-variate distribution involves two attributes.

Attributes

- Attribute Types:
  - Nominal (Related to Names)
  - Binary (Only two states)
  - Numeric: quantitative
    - Interval-scaled
    - Ratio-scaled

Attribute Types

- **Nominal (categorical):** means "relating to names.
- The values of a **nominal attribute** are symbols, *names of things*, categories and states.
  - *Hair_color* = {*auburn, black, blond, brown, grey, red, white*}
  - Marital status, occupation, ID numbers, zip codes etc
- **Binary**
  - Nominal attribute with only 2 states (0 and 1)
- Where 0 typically means that the attribute is absent, and 1 means that it is present.
- Also referred to as **Boolean** if the two states correspond to *true* and *false*.

Attribute Types

- Symmetric binary: Both outcomes equally important, carry the same weight. e.g., gender
- Asymmetric binary: outcomes not equally important.
  - e.g., medical test (positive vs. negative)
  - Convention: assign 1 to most important outcome (e.g., HIV positive)
- **Ordinal**
- Values have a meaningful order (ranking) but magnitude between successive values is not known.
- *Size* = {*small, medium, large*}, grades, army rankings

Numeric Attribute Types

- **Numeric Attributes (integer or real-valued)**
- Interval-scaled
- Ratio-scaled
- **Interval**
  - Measured on a scale of **equal-sized units**
  - Values have order
    - E.g., *temperature in C˚or F˚, calendar dates*
    - *0°C does NOT mean “no temperature”*
      ❌*20°C is NOT twice as hot as 10°C* ❌

Numeric Attribute Types

- **Ratio**
  - Inherent **zero-point (**Zero represents the **complete absence** of the quantity being measured)
  - We can speak of values as being an order of magnitude larger than the unit of measurement (10 K˚ is twice as high as 5 K˚).
  - Weight 0 kg means no weight
  - 10 kg is twice 5 kg

Discrete vs. Continuous Attributes

- **Discrete Attribute**
  - Has only a finite set of values
    - E.g., zip codes, profession, or the set of words in a collection of documents
    - {44000, 54000, 60000} Even though they are numbers, they are:
    - Labels
    - Not continuous quantities
    - Finite possible value
  - Sometimes, represented as integer variables
  - Note: Binary attributes are a special case of discrete attributes: {0,1 } Only two possible values → finite → countable → discrete

Discrete vs. Continuous Attributes

- **Continuous Attribute**
  - Has real numbers as attribute values
    - E.g., temperature, height, or weight
    - 25°C, 25.1°C, 25.12°C, 25.123°C …Between 25 and 26, there are infinitely many possible values.
  - Practically, real values can only be measured and represented using a finite number of digits
  - Continuous attributes are typically represented as floating-point variables

Outline

- Data Objects and Attribute Types
- Basic Statistical Descriptions of Data

Basic Statistical Descriptions of Data

- <u>Motivation</u>
- To identify properties of the data and highlight which data values should be treated as noise or outliers.
  - To better understand the data: central tendency, variation and spread
- <u>Central tendency</u>
- Location of the middle or center of a data distribution
- Mean, median, mode, and midrange

Measuring the Central Tendency

- Mean (algebraic measure) (sample vs. population):
  Note: n is sample size (Number of values selected from the population) and N is population size (Total number of values in the entire population).

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \mu = \frac{\sum x}{N}$$

$$\bar{x} = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

Example: If a class has **100 students (N = 100)**, but you collect marks of **20 students (n = 20)**: Mean of 100 students → divide by **100** Mean of 20 students → divide by **20**

- Weighted arithmetic mean:

Example:

Suppose you have three exam scores: 80, 90, and 70, with weights 20%, 50%, and 30% respectively.

$$\text{Weighted Mean} = \frac{(80 \times 0.2) + (90 \times 0.5) + (70 \times 0.3)}{0.2 + 0.5 + 0.3}$$

$$= \frac{16 + 45 + 21}{1} = 82$$

So the weighted average score is **82**, not the simple average $(80 + 90 + 70)/3 = 80$.

Measuring the Central Tendency

- Median:
  - Middle value if odd number of values, or average of the middle two values otherwise
- Mode
  - Value that occurs most frequently in the data
- Data sets with one, two, or three modes are respectively called **unimodal**, **bimodal**, and **trimodal**.

Number of Modes:
Unimodal: Only one value occurs most frequently.
Bimodal: Two values occur with the same highest frequency.
Trimodal: Three values occur with the same highest frequency.
Multimodal: More than three values occur with the same highest frequency

- Dataset: 3, 5, 3, 8, 9, 3 → Mode = 3 (occurs 3 times)
- Dataset: 2, 4, 4, 6, 6 → Modes = 4 and 6 (bimodal)

Measuring the Central Tendency

- Midrange
  - It can also be used to assess the central tendency of a numeric data set.
  - It is the average of the largest and smallest values in the set.

Example 2.6 **Mean.** Suppose we have the following values for *salary* (in thousands of dollars), shown in increasing order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110. Using Eq. (2.1), we have

$$\bar{x} = \frac{30 + 36 + 47 + 50 + 52 + 52 + 56 + 60 + 63 + 70 + 70 + 110}{12}$$

$$= \frac{696}{12} = 58.$$

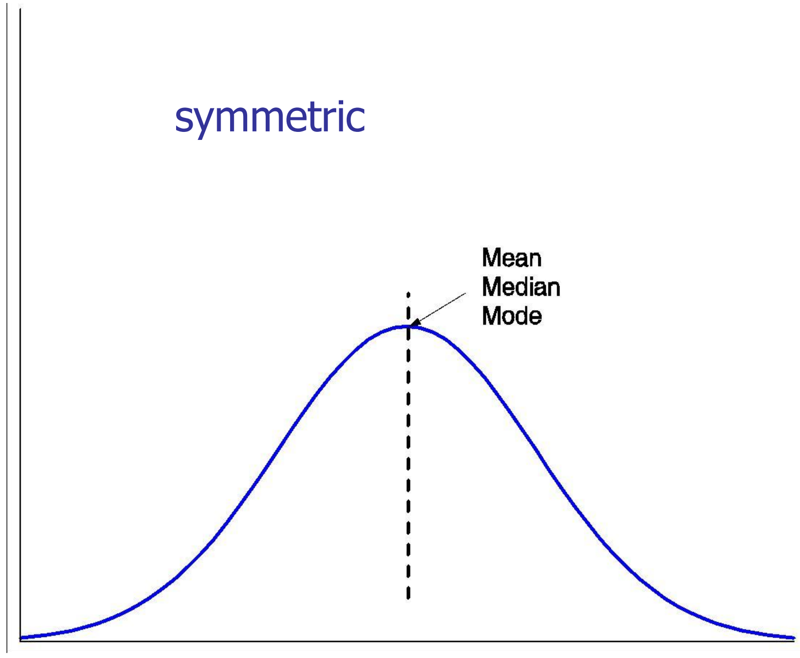
Thus, the mean salary is \$58,000. ∎

Midrange. The midrange of the data of Example 2.6 is $\frac{30{,}000+110{,}000}{2} = \$70{,}000$.

Symmetric vs. Skewed Data

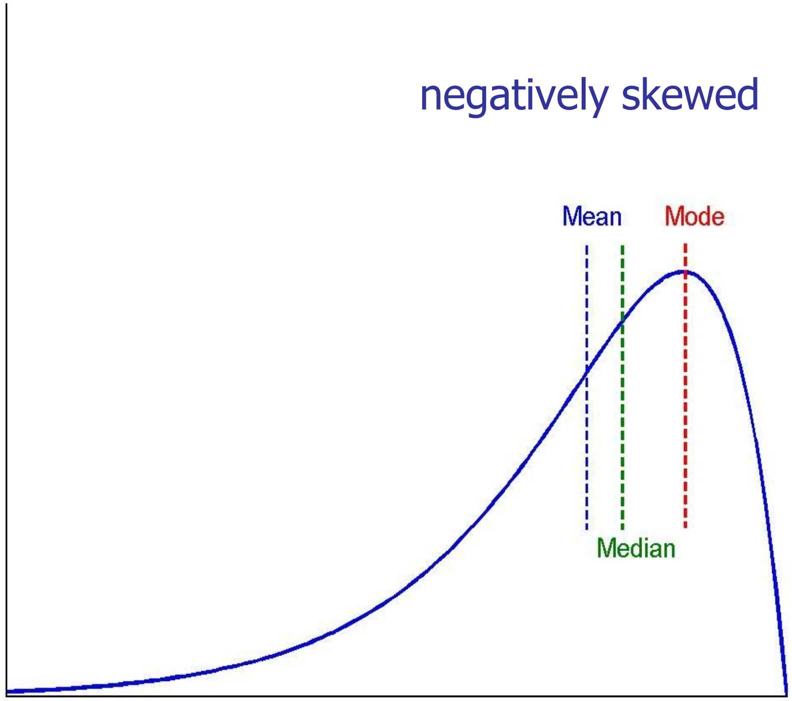
- Median, mean and mode of symmetric, positively and negatively skewed data

- For unimodal frequency curve with perfect **symmetric** data distribution, the mean, median, and mode are all at the same center value



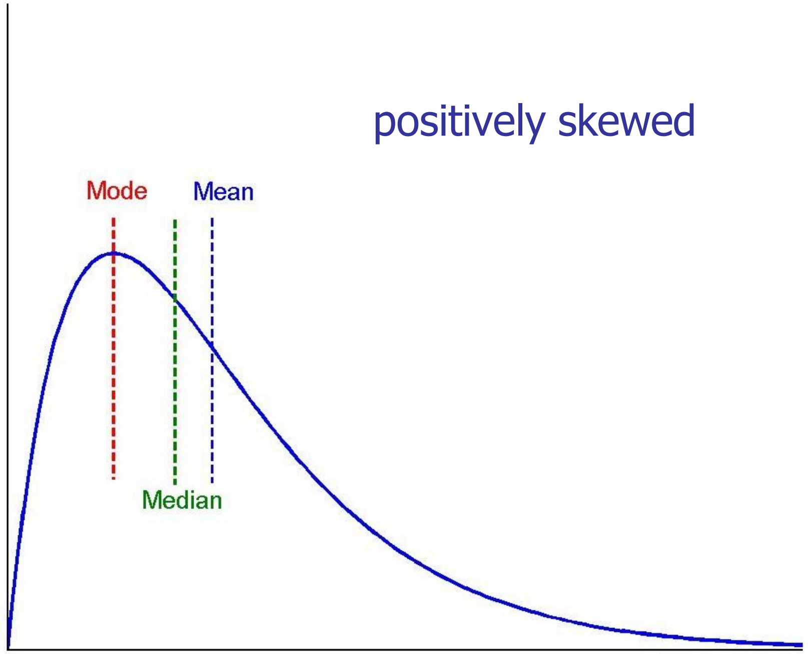
Symmetric vs. Skewed Data

- **For negatively skewed**, the mode occurs at a value greater than the median



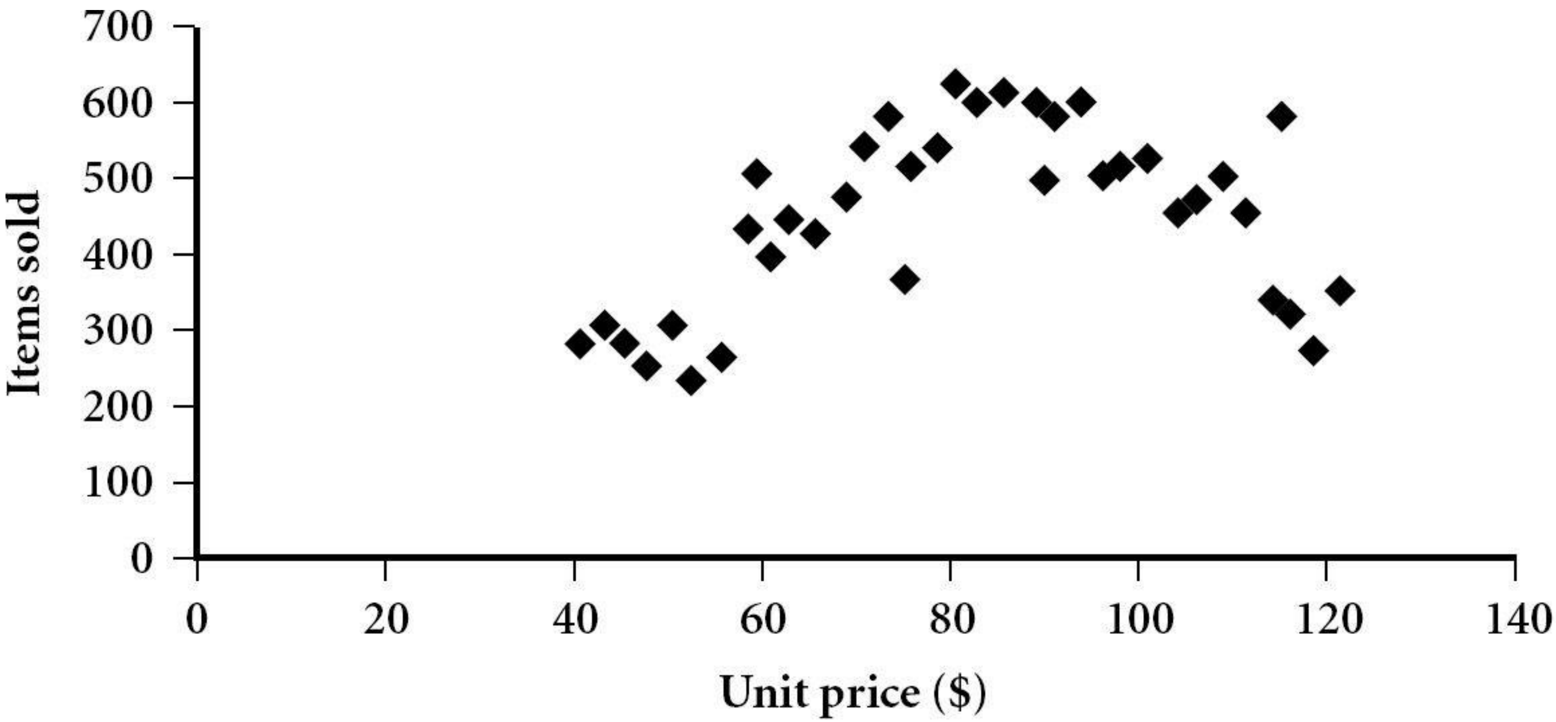
Symmetric vs. Skewed Data

- **For positively skewed**, the mode occurs at a value that is smaller than the median

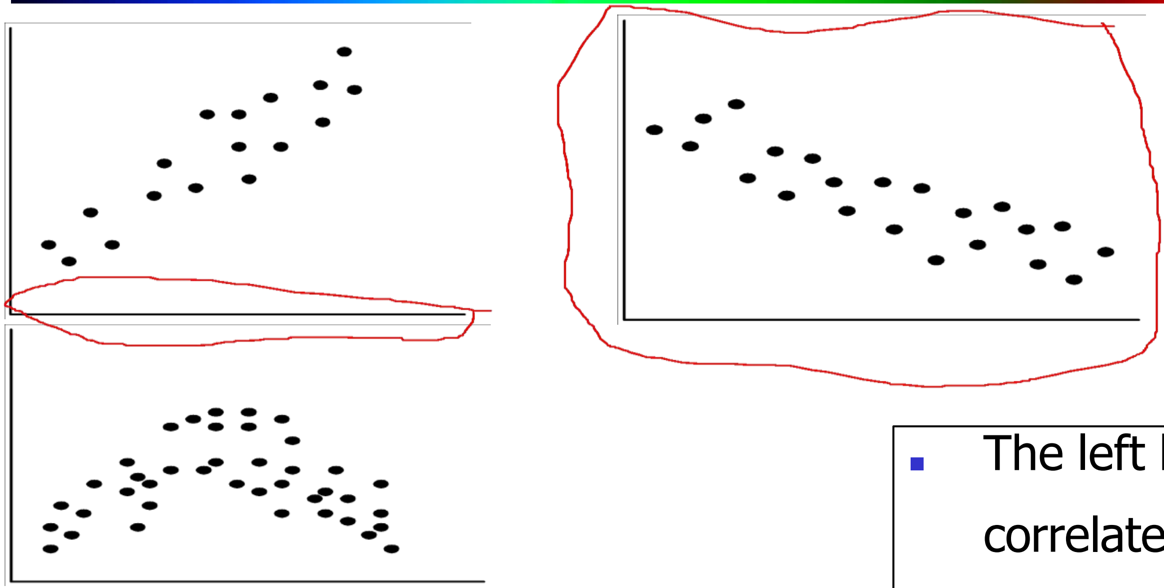


Scatter plot

- Provides a first look at bi-variate data to see clusters of points, outliers, etc
- Each pair of values is treated as a pair of coordinates and plotted as points in the plane



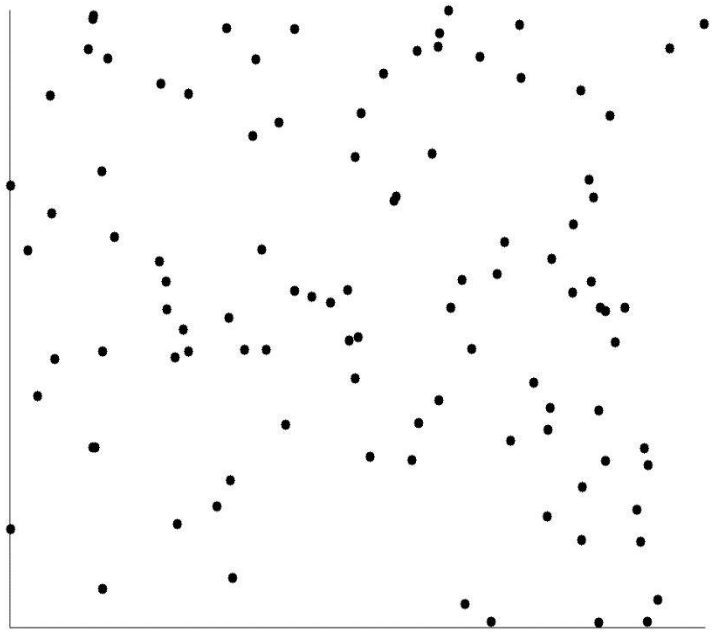
Positively and Negatively Correlated Data

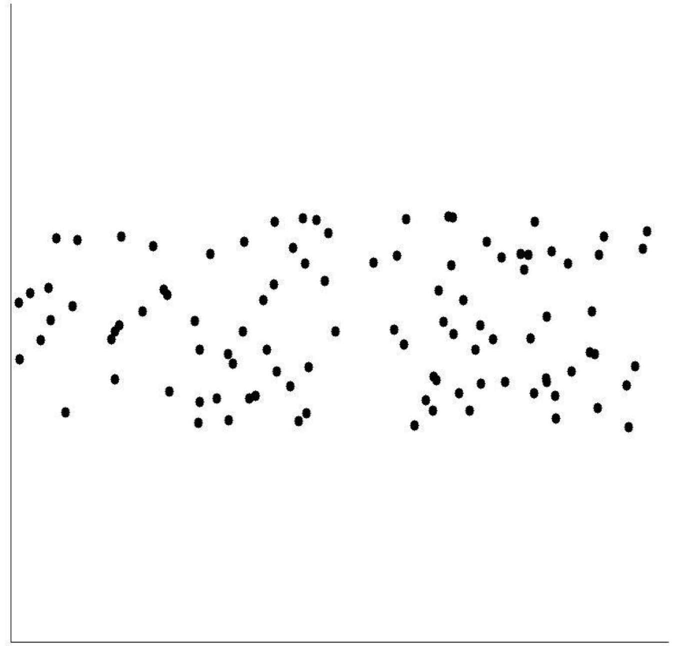


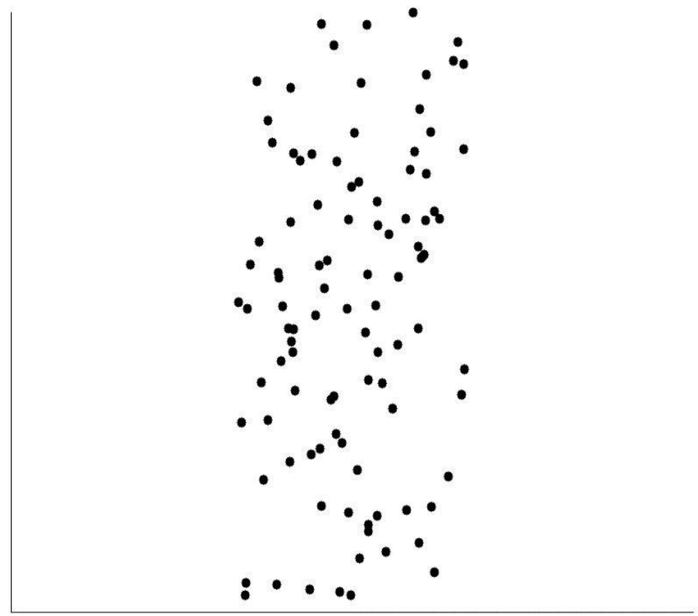
- The left half fragment is positively correlated
- The right half is negative correlated

Type	Relationship	Correlation (r)	Example
Positive Correlation	Both increase or decrease together	0 to +1	Hours studied vs. marks
Negative Correlation	One increases, other decreases	0 to -1	Social media vs. exam marks

Uncorrelated Data







Graph

•Points are scattered **randomly** on a graph, **no upward or downward trend**.

•Even if the points move a little, **no clear linear or non-linear pattern** exists

Any Question